Expectation

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Sep 2026

Demand Forecasting

Motivation

A retailer must decide how many units Q of a product to stock for tomorrow. Demand is uncertain. If demand exceeds inventory, sales are lost. If inventory exceeds demand, unsold units have no salvage value.

The goal is to choose Q to maximize expected profit.

Model Assumptions

Let demand D be uniformly distributed on $[0, 100]$:

$$D \sim \text{Uniform}(0, 100).$$

Thus the density of D is

$$f(d) = \frac{1}{100}, \qquad 0 \leq d \leq 100.$$

Economic parameters:

- Selling price per unit: 100
- Cost per unit: 60
- No salvage value for unsold units

Profit Function

The retailer can only sell $\min(D, Q)$ units. Revenue is therefore:

$$100 \min(D, Q).$$

The total cost is:

$$60Q.$$

Hence profit is:

$$\Pi(Q, D) = 100 \min(D, Q) - 60Q.$$

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Expected Profit

We compute expected profit using:

$$\mathbb{E}[\Pi(Q, D)] = \int_0^{100} \Pi(Q, d) f(d)\, dd.$$

Substituting $f(d) = \frac{1}{100}$:

$$\mathbb{E}[\Pi(Q, D)] = \frac{1}{100} \int_0^{100} [100 \min(d, Q) - 60Q]\, dd.$$

Splitting the Integral

The function $\min(d, Q)$ changes form at $d = Q$.

Case 1: $0 \leq d \leq Q$
In this region:

$$\min(d, Q) = d.$$

Profit becomes:

$$100d - 60Q.$$

Case 2: $Q \leq d \leq 100$
In this region:

$$\min(d, Q) = Q.$$

Profit becomes:

$$100Q - 60Q = 40Q.$$

Thus expected profit becomes:

$$\mathbb{E}[\Pi(Q, D)] = \frac{1}{100} \left[\int_0^{Q} (100d - 60Q)\, dd + \int_Q^{100} 40Q\, dd \right].$$

Evaluating the First Integral

$$\int_0^{Q} 100d\, dd = 100 \cdot \frac{Q^2}{2} = 50Q^2.$$

$$\int_0^{Q} 60Q\, dd = 60Q^2.$$

Therefore,

$$\int_0^{Q} (100d - 60Q)\, dd = 50Q^2 - 60Q^2 = -10Q^2.$$

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Evaluating the Second Integral

$$\int_{Q}^{100} 40Q \, dd = 40Q(100 - Q).$$

Combining the Terms

$$\mathbb{E}[\Pi(Q, D)] = \frac{1}{100} \left[-10Q^2 + 40Q(100 - Q) \right].$$

Expanding:

$$40Q(100 - Q) = 4000Q - 40Q^2.$$

So:

$$= \frac{1}{100} \left[4000Q - 50Q^2 \right].$$

Final Expression for Expected Profit

$$\boxed{\mathbb{E}[\Pi(Q)] = 40Q - 0.5Q^2}$$

Finding the Optimal Order Quantity

Differentiate with respect to Q:

$$\frac{d}{dQ}\mathbb{E}[\Pi(Q)] = 40 - Q.$$

Set derivative equal to zero:

$$40 - Q = 0$$

$$\boxed{Q^* = 40}.$$

Remark: To confirm that its a maxima, one should ideally do the second derivative test as well.

Maximum Expected Profit

$$\mathbb{E}[\Pi(40)] = 40(40) - 0.5(40^2) = 1600 - 800 = \boxed{800}.$$

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